

Volleyball-Group-Activity-Recognition

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Abstract—A condensed summary of the Volleyball Group Activity Recognition project: the data pipeline, the generic multi-baseline loader, and the three completed baselines — B1, a single-frame fine-tuned ResNet-50 (62.6% test accuracy / 0.630 macro F1); B3, a person-then-group crop classifier (60.7% / 0.589); and B4, a temporal classifier that runs an LSTM over B1’s frozen frame features (66.1% / 0.673, the strongest result so far). The dataset (Ibrahim et al., CVPR 2016) contains 55 videos / 4,830 clips with 8 scene-level group activities and 9 player-level actions.

Keywords— Volleyball, Group Activity Recognition, CNN, RNN, LSTM, ResNet50, Classification, Multiclass Classification, Transfer Learning

1. Problem & Dataset

Volleyball Group Activity Recognition is a hierarchical classification problem: predict the scene-level group activity (8 classes: 1/r-pass, -spike, -set, -winpoint) and the per-player action (9 classes: blocking, digging, falling, jumping, moving, setting, spiking, standing, waiting). The dataset comes from “A Hierarchical Deep Temporal Model for Group Activity Recognition” (Ibrahim et al., CVPR 2016; official repo, Kaggle mirror).

Key facts: **55 videos, 4,830 clips** (train/val/test = 2,152/1,341/1,337 clips over disjoint video splits of 24/15/16). Each clip has **41 frames**; the clip folder is named after its annotated *middle frame*. Both label levels are heavily skewed: standing is $\approx 68\%$ of person actions, and 1/r_winpoint are $\approx 2.5\times$ rarer than the other group classes. The group classes come in **left/right mirror pairs**, so horizontal geometry is part of the label — the transform pipelines therefore use no horizontal flips and warp inputs directly to 224×224 so the full court (or the full player) always stays in view.

2. Data Pipeline

Three raw annotation sources (action detections, person detections, player tracking) are unified by a two-stage parser (`src/json_parser.py`) into one master JSON keyed `video_id/clip_id`, then enriched with the group-activity label per clip (looked up per video against each clip’s own middle frame) and cached as a pickle (~ 247 MB) for instant metadata loading. Frames are served either from a memory-mapped LMDB (local) or straight from disk (Kaggle); both backends expose the identical `VolleyballDataset`, which covers every baseline through flags: full frames or per-player crops, 1 or 9 frames per clip. A custom collate pads the variable player count per clip and returns a validity mask so padded slots never reach the loss or the pooling.

3. Baseline Roadmap

Table 1 lists the planned progression of models, each adding temporal and/or player-level structure on top of the previous one.

Table 1. Baseline models. B1, B3, and B4 are complete; B5–B8 pending.

Model	Temporal	Player	Scene
B1	—	—	Image clf.
B3	—	Crop clf.	Concat-pool + MLP
B4	LSTM	—	LSTM
B5	LSTM/player	Max-pool	NN
B6	LSTM/image	Max-pool	LSTM
B7	$LSTM_1 + LSTM_2$	Max-pool	$LSTM_2$
B8	$LSTM_1 + LSTM_2$	Team-split	$LSTM_2$

4. Baseline 1: Single-Frame Classifier

Fine-tunes a pretrained ResNet-50 on the clip’s middle frame for the 8 group classes. Two-stage schedule: linear probe (head only, 10 epochs, $lr\ 10^{-3}$), then partial fine-tune of `layer3/4 + head` (up to 100 epochs, backbone $lr\ 2.5 \times 10^{-4}$, head $3\times$, cosine annealing, batch 16, frozen BatchNorm held in eval).

Table 2. Baseline 1 test-set metrics.

Metric	Value
Accuracy	62.60%
Macro F1	0.630
Test Loss	1.42

Fig. 1 shows the per-class confusion on the test split.

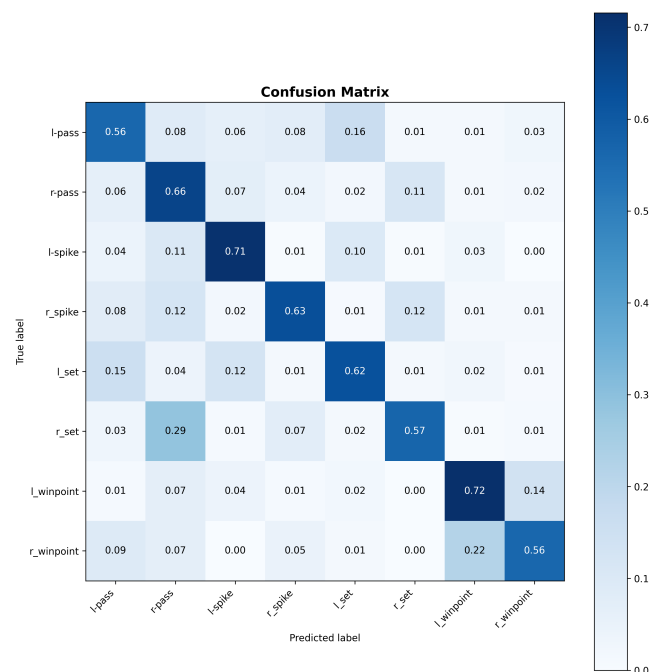


Figure 1. Baseline 1 test confusion matrix.

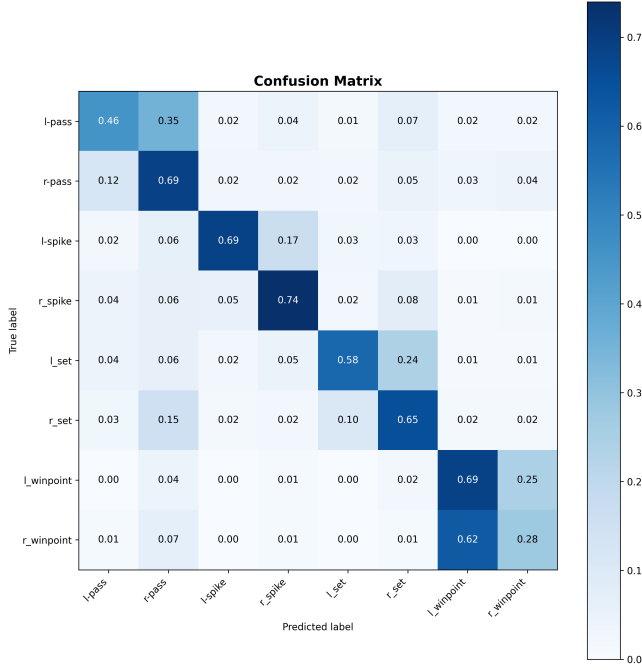
5. Baseline 3: Person-Then-Group

Stage A fine-tunes ResNet-50 on individual player crops for the 9 person actions with inverse-frequency class-weighted CE (without it the loss collapses onto `standing`). **Stage B** freezes that backbone, extracts a $(P, 2048)$ feature matrix per clip, concatenates masked max- and mean-pooling across players (2×2048), and trains an MLP ($4096 \rightarrow 512 \rightarrow 8$, Dropout 0.4) with class-weighted CE for the group classes. SGD (momentum 0.9, Nesterov), 50 epochs and patience 10 per stage, batch 16 clips, `nn.DataParallel` on multi-GPU hosts. Stage A reaches 70.4% validation accuracy (macro F1 0.512) on the person actions; Stage B 60.0% (0.584) on the group activities.

Fig. 2 shows the dominant error mode: left/right mirror confusion within each activity pair, worst on the rare `r_winpoint`. The main open issue is **overfitting** (train $\approx 95\%$ vs. val 58–70%), to be addressed with stronger augmentation, higher dropout, and earlier stopping in a follow-up run.

Table 3. Baseline 3 test-set metrics.

Metric	Value
Accuracy	60.73%
Macro F1	0.589
Test Loss	1.09

**Figure 2.** Baseline 3 test confusion matrix.

6. Baseline 4: Temporal Image Classifier

Each clip’s 9-frame window is pushed through a **frozen** feature extractor loaded from Baseline 1’s fine-tuned backbone, producing a (9, 2048) feature sequence per clip. A single-layer LSTM (hidden 512) consumes the sequence and its final hidden state is classified by a small MLP head (512 $\rightarrow$ 256 $\rightarrow$ 8, Dropout 0.3). Only the LSTM + head train (≈ 5.4 M parameters; SGD lr 10^{-3} , cosine annealing, class-weighted CE, batch 16 clips).

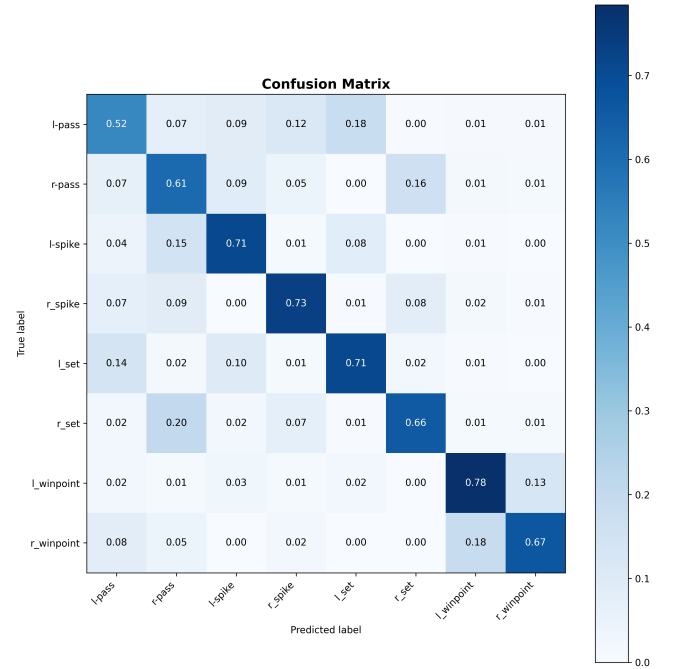
Table 4. Baseline 4 test-set metrics.

Metric	Value
Accuracy	66.12%
Macro F1	0.673
Test Loss	1.05

B4 is the strongest baseline so far: temporal context over B1’s own features is worth **+3.5 accuracy points / +0.043 macro F1** versus B1’s single-frame result — the first direct evidence in this project that motion carries group-activity signal. Convergence is very fast (best validation F1 at epoch 4; training accuracy $\approx 99\%$ by epoch 14, where early stopping fired), repeating the overfitting pattern of B1/B3. Fig. 3 shows the test confusion matrix.

7. Next Steps

(i) Reduce overfitting across all baselines — B4 peaks at epoch 4, so stronger augmentation, higher dropout, and earlier stopping are the levers; (ii) proceed to the temporal crop baselines B5–B8 — noting that player tracking covers only $\approx$ half of each clip’s frames, so those baselines will exercise the loader’s detection-fallback path, which should be reviewed before B5.

**Figure 3.** Baseline 4 test confusion matrix.